Mst Afsana Mimi

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Summary

Statistics master's graduate with experience transforming large datasets into reliable analysis, recurring reports, and clear quantitative findings. Proficient in Excel, SQL, Python, R, and Tableau, with strengths in performance monitoring, dashboard development, data validation, statistical analysis, and data storytelling. Brings curiosity, attention to detail, and clear communication to collaborative, deadline-driven work.

Education

University of Massachusetts Amherst
Master's in Statistics (CGPA: 3.77) Sep 2023 – May 2026

University of Dhaka
Master's in Applied Statistics (CGPA: 3.75) Jan 2017 – Dec 2018

University of Dhaka
Bachelor's in Applied Statistics (CGPA: 3.61) Jan 2013 – Dec 2016

Technical Skills

Analytics & Reporting

- Excel (pivot tables, lookups, charts), SQL (joins, aggregations, window functions), Python (Pandas, NumPy), and R
- Tableau, dashboard development, recurring reporting, KPI monitoring, data visualization, and data storytelling

Data Analysis & Research

- Data collection, cleaning, transformation, integration, validation, and quality checks
- Trend and anomaly analysis, statistical modeling, hypothesis testing, measurement planning, and evaluation
- Primary and secondary research support; translating findings into actionable recommendations

Workflows & AI Tools

- Git/GitHub, reproducible analytical workflows, and AWS familiarity
- Prompt-based use of generative AI tools, including ChatGPT and Gemini, for research, ideation, and workflow support

Experience

Teaching Assistant (Data & Statistics)
UMass Amherst Sep 2023 – May 2026

- Supported R- and Python-based analysis projects while balancing recurring responsibilities and academic deadlines.
- Explained statistical results clearly and helped translate quantitative findings into defensible conclusions.

Research Assistant (NLP & Data Analysis)
University of Dhaka Jul 2021 – Jun 2022

- Processed large-scale text datasets and built Python workflows for ingestion, cleaning, transformation, and feature engineering.
- Developed and evaluated machine-learning models, monitored performance metrics, and summarized findings for the research team.
- Performed validation and consistency checks to maintain accurate, analysis-ready datasets.

Research Assistant (Socio-economic Data Analysis)
University of Dhaka Jul 2018 – Jun 2019

- Cleaned, transformed, and structured large-scale survey datasets for analysis.
- Built reproducible analytical workflows and applied statistical models to identify meaningful socio-economic trends.
- Communicated findings to support data-driven policy analysis and maintained collaborative code using Git.

Projects

Can Agentic AI Eliminate the Need for a Dedicated Relational Foundation Model?

- Developing an agentic machine-learning framework for relational databases to automate schema understanding, relational evidence construction, and predictive reasoning over structured data.
- The long-term objective is to build a benchmark for evaluating agentic systems on realistic database tasks.

Gaussian Process Modeling with Fairness Constraints

- Developing Gaussian Process models with fairness constraints across demographic groups.
- Designing evaluation strategies to assess trade-offs between predictive accuracy and bias.

High-Dimensional Feature Selection (MCAR)

- Developed an MCAR feature-selection method for censored high-dimensional data using AFT models.
- Evaluated predictive performance through simulations and DLBCL microarray data.

Awards and Scholarship

- Conference Awards for Scientists (CASc) by International Society for Clinical Biostatistics (ISCB).
- National Science and Technology (NST) Fellowship Award by Education Board of Bangladesh.
- Dhaka University Merit Scholarship for academic results in B.S. Hon's examination.

Publications

- Mimi, A., & Khan, M. H. R. (2021). Variable selection for censored data using modified correlation adjusted correlation (mcar) scores. Statistics in Medicine, 40(23), 5046–5064.

Google Scholar: https://scholar.google.com/citations?user=LaVyjwYAAAAJ&hl=en&oi=sra